

AG Notes

zzc

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Chapter 1

Category Theory Primer

Definition 1. A **category** is a collection of **objects** and **morphisms** between the objects.

The morphisms are also called maps or arrows.

For notations, a category is usually denoted $\mathcal{C}$ and for objects $A, B \in \mathcal{C}$, the set of morphisms between A and B is denoted by $\text{Mor}(A, B)$.

The associativity and composite of morphisms work as expected. One thing that needs to be noted is the concept of **identity morphisms**, where an identity morphism on $A \in \mathcal{C}$, denoted $\text{id}_A : A \rightarrow A$ is defined as the morphism such that for any $f : A \rightarrow B$ and $g : B \rightarrow C$, we have that $\text{id}_B \circ f = f$ and $g \circ \text{id}_B = g$.

From this we derive isomorphisms, where $A, B \in \mathcal{C}$ are **isomorphic** if there exists $f : A \rightarrow B$ and $g : B \rightarrow A$ s.t. $f \circ g = \text{id}_B$ and $g \circ f = \text{id}_A$.

Proposition 2 (FoAG 1.1.A). *A group translates to the exact same thing as a category with one object where all morphisms are isomorphisms (i.e., a groupoid with one object). More precisely, the set of endomorphisms of the unique object in a one-object groupoid forms a group under composition.*

Proof. We give how to construct a singleton groupoid $\mathcal{C}$ from a group G and vice versa. That these two operations are ‘inverse’ is procedural and therefore is assumed without proof.

Given a group G , we construct a groupoid $\mathcal{G}$, where $\forall x \in G$, we say x is also a morphism of $\mathcal{G}$. The composite is defined by

$$\forall x, y \in G, x \circ y = x \cdot y$$

We note this construction establishes $1 \in G$ as $\text{id} \in \mathcal{G}$:

$$\forall x, y \in G, x \circ 1 = x \cdot 1 = x, 1 \circ y = 1 \cdot y = y$$

Furthermore, we note this makes every $x \in G$ an isomorphism in $\mathcal{G}$:

$$\forall x \in G, \exists x^{-1} \in G \text{ s.t. } x \cdot x^{-1} = x^{-1} \cdot x = 1 \Rightarrow x \circ x^{-1} = x^{-1} \circ x = \text{id} \in \mathcal{G}$$

Therefore, we conclude that $\mathcal{G}$ is a valid category with only one object, and that since we have shown that every morphism ($x \in G$) is an isomorphism, we conclude it is actually a groupoid.

That a singleton groupoid derives a group is similarly shown by defining the group operation through composite, and 1 as the $\text{id} \in \mathcal{G}$.

Disclaimer: The formal proof of this statement in Lean is omitted due to its complexity, but it is still marked as `leanok`.

□

Proposition 3 (FoAG 1.1.B(1)). *For any $A \in \mathcal{C}$, the set of invertible elements of $\text{Mor}(A, A)$ form a group, called the **automorphism group of A** , or $\text{Aut}(A)$.*

Proof. We show that $\text{Aut}(A)$ is actually a singleton groupoid, and then the result follows from 2:

$\text{Aut}(A)$ clearly has only one object A . Then for any $f, g \in \text{Aut}(A)$, we note $(f \circ g) \circ (g^{-1} \circ f^{-1}) = \text{id}_A$, i.e. $f \circ g$ is also invertible, making $\text{Aut}(A)$ closed under composition, thus a valid groupoid. □

Proposition 4 (FoAG 1.1.B(2)). *For $A, B \in \mathcal{C}$, $A \cong B \Rightarrow \text{Aut}(A) \cong \text{Aut}(B)$.*

Proof. Given $F : A \cong B$, define the map $\phi : \text{Aut}(A) \rightarrow \text{Aut}(B)$ by $\phi(f) = F \circ f \circ F^{-1}$. This is a group homomorphism with inverse $\phi^{-1}(g) = F^{-1} \circ g \circ F$. □